Geo-Insight

A Bayesian Latent-Variable Model
for Ranking Overlooked Humanitarian Crises

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Abstract

In 2026, agencies are asking \$33B to reach 135 M people; the 2025 appeal raised \$12B, the lowest in a decade, with 19 contexts below 50 % coverage. When nearly every response is underfunded, a single coverage ratio stops discriminating. We treat overlookedness as a latent scalar θ — never observed directly — and infer a posterior over θ for each humanitarian crisis from six noisy signals through a hierarchical Bayesian observation model. Each signal is linked to θ through a likelihood matched to its support (beta regression for bounded fractions, log-normal for the per-PIN gap, ordered logistic for INFORM severity); slopes are sign-constrained by domain knowledge; a population-level prior pools partial-data countries toward the global mean. Stakeholder disagreement is encoded as a difference between priors over the six attribute slopes — not weights in a sum — yielding a separate posterior per stakeholder and a principled measure of consensus from posterior overlap. Inference is variational with NUTS validation. The candidate pool is restricted to HRP-eligible countries (those with an active humanitarian response plan), which is the population CERF UFE picks from. Implemented over a typed semantic layer of 71 properties across five derivation levels. External validation against two independent human-curated benchmarks: 5/10 precision @ 10 on CERF UFE 2025 w1 (tied with the additive baseline), 2/7 on CERF UFE 2025 w2 (vs 1/7), 3/10 on CARE *Breaking the Silence* 2024 (vs 1/10) — with calibrated uncertainty (Spearman $\rho = 0.89$ between SVI medians and the NUTS gold standard).

1 Motivation

The 2026 Global Humanitarian Overview asks \$33B for 135 M people, with a hyper-prioritised \$23B floor targeting 87 M lives.[1] The 2025 appeal raised \$12B. Coverage ratios have collapsed — DRC ∼22 %, Yemen and Somalia ∼24 %, Sudan from 69 % (2024) to 35 % (2025). Nineteen contexts are flagged as severely underfunded, up from eight in 2021. A single coverage ratio cannot distinguish between 19 countries that are all below 50 % coverage; the useful variation is no longer *whether* a crisis is underfunded but *how*, and *for whom*.

Three kinds of overlookedness become distinguishable once scoring moves from scalar to joint distribution: *forgotten* (low funding, low attention — Angola drought 2023), *contested* (high attention, persistent decline — Sudan 2025), and *sector-starved* (adequate aggregate, one sector at 20 %). Each warrants a different response — multi-year pooled funding, targeted advocacy, sector-level reallocation — and current tools conflate them.

2 Related Work

Rye & Aktas [1] introduce an additive multi-attribute framework whose Table 13 demonstrates that ranks diverge under different decision-maker weightings; we treat that finding as motivation

[1] https://www.unocha.org/publications/report/world/global-humanitarian-overview-2026-enesfr

to model the underlying construct as latent and the stakeholders as priors, rather than to pick weights. Three papers converge on equity as the open gap. Abdulrashid et al. [2] (2026 review, §7.3) list equity-aware prioritisation as one of only five open research items. Sahebjamnia et al. [3] call for *"egalitarian and utilitarian measures"*. Vargas Florez et al. [4] give the equation template (Eqs. 9–12) for fair-distribution penalties; we generalise from inter-regional to inter-cluster inequality within a single country response. Griffith et al. [5] argue for interpretable analytics in time-constrained humanitarian operations, which we take as a hard constraint: every added term must decompose cleanly and be explainable to a non-technical analyst. The Bayesian-latent-variable formulation we adopt here is standard machinery (item-response models, factor analysis with sign-constrained loadings, partial-pooling hierarchical models); the contribution is its application to overlookedness with stakeholders as priors over slopes and external validation against CERF UFE.

3 Data

Six upstream datasets, with declared provenance. Five from OCHA: HNO (people in need per country and sector), HRP (plan metadata), FTS (requirements, funding received, incoming flows per donor), CoD-PS (population), and CBPF (pooled-fund allocations and donor contributions). One from the EU Joint Research Centre: INFORM Severity, as 67 monthly global snapshots consolidated into long-format CSVs (109 countries, 1–5 ordinal severity plus sub-indicators: affected, displaced, fatalities, phase-level PIN splits, access-limited/restricted populations). Two externally-curated lists for validation only, not training: CERF Underfunded Emergencies (22 country selections across 2024–25) and CARE *Breaking the Silence* (annual top-10 by media attention, not funding). Countries missing more than half of the six scoring signals are withheld rather than imputed.

4 Methodology

Unit of analysis is country-year. A country is *eligible* if it has an active humanitarian response plan, operationalised as: an observed gap-per-person-in-need (a_2). This restriction matches the population CERF Underfunded Emergencies allocations are drawn from, so external validation is on the same pool. Countries without an HRP are returned with no posterior — the construct "overlooked humanitarian crisis" is not well-defined for them, and their absence from the ranking is a feature, not a bug.

Six-attribute vector per eligible country-year:

- a_1 coverage shortfall $= 1 - \min(\text{fund}/\text{req}, 1) \in [0, 1]$
- a_2 gap per PIN $= \max(0, \text{req} - \text{fund})/\text{PIN} \in [0, \infty)$
- a_3 need intensity $= \text{PIN}/\text{population} \in [0, 1]$
- a_4 severity: INFORM 1–5 ordinal
- a_5 donor concentration (Herfindahl over FTS single-destination incoming flows) $\in [0, 1]$
- a_6 intra-crisis equity: PIN-weighted Gini over sector coverage ratios $\in [0, 1]$,

$$a_6(u) = \frac{\sum_{i,j} n_i n_j |k_i - k_j|}{2\,(\sum_c n_c)^2\,\bar{k}_w}, \quad \bar{k}_w = \frac{\sum_c n_c k_c}{\sum_c n_c}.$$

Attributes are left on their natural support; no min–max normalisation is applied. The observation likelihoods below operate on raw scales and absorb dispersion through their own dispersion parameters.

Latent variable. For each eligible country u, overlookedness is a latent scalar $\theta(u) \in \mathbb{R}$. Larger θ means more overlooked. We never observe θ directly; we observe the six attributes through the likelihoods below.

Observation model. For each attribute i, a likelihood matched to its support:

$$a_i(u) \mid \theta(u) \;\sim\; \begin{cases} \mathrm{Beta}\big(\sigma(\alpha_i + \beta_i\theta)\,\phi_i,\ (1-\sigma(\alpha_i+\beta_i\theta))\,\phi_i\big) & i \in \{1,3,5,6\} \\ \mathrm{LogNormal}(\alpha_2 + \beta_2\theta,\ \sigma_2^2) & i = 2 \\ \mathrm{OrderedLogistic}(\alpha_4 + \beta_4\theta,\ \kappa) & i = 4 \end{cases} \tag{1}$$

where $\sigma(\cdot)$ is the logistic function, $\phi_i > 0$ is a per-attribute precision, $\sigma_2 > 0$ is the log-normal scale, and $\kappa = (\kappa_1, \kappa_2, \kappa_3, \kappa_4)$ are the ordered-logistic cut-points. Beta regression respects the bounded support of the four fractional attributes; the log-normal handles the positive, multiplicative-scale per-PIN gap; the ordered logistic handles the qualitative spacing of INFORM severity steps that an interval-scale treatment would silently distort.

Sign constraints and identification. Every slope is constrained $\beta_i > 0$ via a half-normal prior. This is the operational definition of θ: the latent moves with attributes the way domain knowledge says it should — higher coverage shortfall, higher per-PIN gap, more concentrated donors, sharper sector imbalance, greater need intensity, higher severity all correspond to greater overlookedness. Without sign constraints the model is only weakly identified and the inferred θ picks up arbitrary linear combinations. The location ambiguity ($\theta \mapsto \theta + c$, $\alpha_i \mapsto \alpha_i - \beta_i c$) is resolved by the population-level prior below, which centres $\mathbb{E}_u[\theta(u)]$ near zero.

Hierarchical prior on θ. A Gaussian population-level prior pools countries:

$$\mu_\theta \sim \mathcal{N}(0, 0.3), \qquad \sigma_\theta \sim \text{Half-}\mathcal{N}(0.5), \qquad \theta(u) \mid \mu_\theta, \sigma_\theta \sim \mathcal{N}(\mu_\theta, \sigma_\theta^2). \tag{2}$$

Countries with sparse data are pulled toward the population mean by the prior; countries with dense data are barely affected. Sparse-data countries get wider posteriors structurally — not because we down-weight them by hand, but because the likelihood has fewer terms to sharpen the posterior on θ.

Stakeholders as priors over slopes. A stakeholder s is encoded as a prior on the six slopes,

$$\beta_i^{(s)} \sim \text{Half-}\mathcal{N}(m_i^{(s)},\ v_i^{(s)}), \qquad i = 1, \ldots, 6, \tag{3}$$

producing a separate posterior $p(\theta \mid D, s)$ per stakeholder. Consensus and contested crises are read directly from posterior overlap across stakeholders, not from rank variance over arbitrary weight vectors. Four stakeholder priors (CERF, ECHO, USAID, NGO) are pre-registered; under the diffuse-prior consistency regime used for the headline result below, all four reduce to the same default.

Inference. Variational, with NUTS validation on every fit. The production guide is `AutoMultivariateNormal` (mean-field is too aggressive — it underestimates posterior variance by $\sim$3$\times$ on this model, with Spearman $\rho = 0.68$ on theta medians against NUTS; the multivariate guide captures the latent–slope correlations and recovers $\rho = 0.89$). `init_to_median`; Adam, learning rate 3×10^{-3}, 8,000 SVI steps; full run $\sim$3 s on CPU. NUTS reference: 1,000 warmup + 2,000 samples $\times$ 4 chains, $\sim$24 s on the same machine. Reported credible intervals are SVI estimates; the SVI-vs-NUTS calibration ratio is noted with every reported result.

Outputs. Per country: posterior median $\widehat{\theta}(u)$, 5th and 95th posterior percentiles forming a 90 % credible interval, CI width as a direct uncertainty proxy. Rankings come from the posterior median; pairs of countries with overlapping CIs are flagged as not differentiable from the data alone.

Four-cell typology. A median split on the posterior CI width against a median split on intra-crisis Gini (a_6) produces the four-cell typology in Table 1.

	Narrow posterior	**Wide posterior**
Sectors uneven	consensus sector-starved	contested sector-starved
Sectors even	consensus overlooked	contested balanced

Table 1: Four-cell typology from posterior CI width × cluster Gini.

Temporal extension. An AR(1) process per country allows $\theta(u,t) = \mu_u + \rho_u(\theta(u,t-1) - \mu_u) + \eta_u(t)$ with $\eta_u \sim \mathcal{N}(0, \tau_u^2)$, where μ_u is the chronic baseline and $\theta(u,t) - \mu_u$ is the acute deviation. Both fall out of the same posterior. Full temporal validation against CERF UFE 2025 windows is the immediate next step (§9); the present paper reports the cross-sectional fit.

5 Typed Semantic Layer

Every analytic quantity is a named property registered in a single specification file. Seventy-one properties at five levels: **L1 Observations** (22 raw primitives), **L2 Ratios** (18 scale-normalised signals), **L3 Inequality** (10 distribution indices: HHI, Gini, concentration measures), **L4 Trends** (12 windowed aggregates: baselines, deltas, trend slopes, persistence), **L5 Posterior** (the Bayesian posterior over θ: median, 5th and 95th percentiles, CI width, completeness, plus the INFORM composite). Each property declares its formula, source dataset, input lineage, unit, granularity, and known failure modes. No analytic column exists in code without a registered declaration; a registry class reads the specification at import time so every displayed value carries its provenance. Eight lenses group subsets of the 71 properties by analytic question; six view modes render each lens (ranked atlas, PCA, clustering, per-country profile, cross-lens radar, validation). All 48 lens-by-mode cells render without per-cell code.

6 Validation

Two human-curated external benchmarks, neither used as training data. CERF UFE — the UN Emergency Relief Coordinator's twice-yearly allocations to underfunded crises — maps to the underfunding-and-consensus axis. CARE BTS (Breaking the Silence, annual top-10) maps to the under-reporting axis. Both are produced by methodology entirely independent of anything in this paper.

The Bayesian posterior is scored against three CERF UFE windows and the most recent CARE BTS list. As an ablation, we also score an additive baseline scorer ($G(u) = \frac{1}{6}\sum_i \tilde{a}_i(u)$, evidence-bounded over observed attributes) on the same pool.

Benchmark	k	**Bayesian**	**Additive baseline**
CERF UFE 2024 w2	10	3/10 (0.30)	3/10 (0.30)
CERF UFE 2025 w1	10	**5/10** (0.50)	**5/10** (0.50)
CERF UFE 2025 w2	7	**2/7** (0.29)	1/7 (0.14)
CARE BTS 2024	10	**3/10** (0.30)	1/10 (0.10)

Table 2: Precision @ k against external benchmarks. Candidate pool: 22 HRP-eligible countries (those with an observed per-PIN gap). The Bayesian model ties the additive baseline on the two larger CERF windows and beats it on the smaller windows where the baseline's variance is most exposed — with a 3× advantage on CARE BTS where the additive baseline collapses. All Bayesian numbers use the production AutoMultivariateNormal SVI guide, calibrated against NUTS (Spearman $\rho = 0.89$ on posterior medians; CI widths within 2× of NUTS).

Calibration. The variational posterior is validated against NUTS, the gold-standard MCMC reference, on every fit. AutoMultivariateNormal SVI runs in $\sim$3 s and recovers NUTS posterior medians with Spearman $\rho = 0.89$; the simpler mean-field AutoNormal guide gives $\rho = 0.68$ and underestimates posterior variance by $\sim 3\times$. The MVN guide is shipped as the production fit; reported credible intervals are SVI estimates with the calibration ratio against NUTS noted.

What the comparison says. The additive scorer gets to the right answer on the most-prominent half of CERF's selections (the HRP big-name crises) but loses systematically on the CARE list and on the secondary CERF windows that surface less-prominent crises. The Bayesian model recovers these because the population-level prior pools partial-data countries toward the global mean rather than letting one extreme observation dominate, and because the sign-constrained slopes lock the latent's interpretation rather than learning an arbitrary linear combination.

Posterior predictive checks. For each of the six attributes, 1,000 replicates are drawn from the fitted posterior and compared against the observed marginal (run via `analysis/bayesian/ppc.py`; figure available at `proposal/figures/fig_ppc.png`). Coverage of the 90 % predictive interval is ≥ 0.91 for every attribute (target 0.90) — the model's marginals are well-calibrated. Per-country Pearson correlation between predicted mean and observed value is high for `coverage_shortfall` (0.76) and `donor_hhi` (0.82), and weak for `need_intensity`, `cluster_gini`, and `severity_category` — attributes that sit near their attribute ceilings (most HRP countries have PIN $\approx$ population and severity $\in \{4, 5\}$) and so carry little cross-country variation for the latent to recover. The latent is best read as a *funding-overlookedness* axis; the ceiling-saturating attributes constrain the posterior but do not drive its discrimination.

7 Results

Table 3 lists the ten countries ranked most overlooked in 2025 under the HRP-eligible pool ($n = 22$). Posterior medians are reported with their 90 % credible intervals; pairs of countries with overlapping CIs cannot be reliably ordered from the data alone.

#	Country	ISO3	$\widehat{\theta}$	CI low	CI high	c
1	Honduras	HND	−0.107	−0.369	+0.183	1.00
2	El Salvador	SLV	−0.241	−0.513	+0.016	1.00
3	Mozambique	MOZ	−0.252	−0.507	+0.013	1.00
4	Somalia	SOM	−0.262	−0.517	−0.011	1.00
5	Guatemala	GTM	−0.267	−0.554	+0.026	1.00
6	Niger	NER	−0.267	−0.516	−0.015	1.00
7	Haiti	HTI	−0.274	−0.535	−0.010	1.00
8	Cameroon	CMR	−0.286	−0.529	−0.019	1.00
9	Venezuela	VEN	−0.287	−0.526	−0.034	1.00
10	Chad	TCD	−0.304	−0.546	−0.066	1.00

Table 3: Top-10 most overlooked, 2025 (HRP-eligible pool, $n = 22$). Higher $\widehat{\theta}$ is more overlooked; the latent is centred near zero by the population prior. CIs are calibrated against NUTS and substantially overlap across the top-10 — set membership is the defensible claim, not strict ordering within it. HND's CI even includes 0, indicating the data alone do not establish that Honduras is more overlooked than the population mean; expert priors (the stakeholder posteriors of §9) sharpen this.

8 Limitations

Stated in full, as a first-class part of the method.

- **Incompleteness is structural, not imputed.** Sparse-data countries get wider posteriors because their likelihood has fewer terms; the population-level prior pools them toward the global mean. No attribute is ever filled in.
- **The candidate pool is HRP-eligible only ($n = 22$).** Countries without a humanitarian response plan return no posterior. This matches CERF UFE's own scope; it is also a deliberate scientific choice — "overlooked humanitarian crisis" is not well-defined for a country that has no documented response operation.
- **Variational posterior may underestimate variance.** The mean-field Gaussian guide is known to under-cover when latent parameters are correlated. NUTS validation on the full posterior takes ∼15 s and is the calibration reference; reported CI widths are SVI estimates.
- **Sign constraints are domain-knowledge commitments.** All six slopes are constrained $\beta_i > 0$. A new attribute that does not co-vary positively with overlookedness would need an opposite-sign prior or a different observation likelihood.
- **Cluster taxonomy drifts** between HNO and FTS. A simple normalisation (lower-case + non-alphanumeric → underscore) is used; a curated cluster map is the next step.
- **INFORM methodology rescaling**, February 2026, makes severity indices pre- and post-boundary not comparable. The ontology declares this as a failure mode; temporal aggregates that span the boundary use the stable ordinal category, not the continuous severity score.
- **Posterior medians without their CIs are misleading.** Pairs of countries with overlapping CIs cannot be reliably ordered from the data alone. Every reported ranking carries the CI as a column.
- **Not automation.** The system ranks; humans decide. Every displayed value carries its provenance.

9 Planned Next Step: A Learned Layer

Three immediate extensions, ranked by maturity.

1. Temporal AR(1). Equation 1 is cross-sectional. The natural next layer is the AR(1) process per country sketched in §4: $\theta(u,t) = \mu_u + \rho_u(\theta(u,t-1) - \mu_u) + \eta_u(t)$. This unlocks a held-out test that genuinely matters operationally — fit on data through 2024, predict CERF UFE 2025 selections — and gives chronic (μ_u) and acute ($\theta - \mu_u$) decompositions for free from the same posterior.

2. Stakeholder posteriors. The slope priors in equation 3 are pre-registered for CERF, ECHO, USAID, and NGO consortia, but the present results use the diffuse-prior consistency regime in which all four reduce to the same default. Fitting four separate stakeholder posteriors and measuring overlap turns the consensus-vs-contested typology from a methodological commitment into an empirical readout.

3. Learned representation layer. A richer view of a crisis is embedded in INFORM's eighteen sub-indicators on displacement, casualties, access, and phase-level need rather than the six aggregates we currently derive. A learned encoder (a sequence model such as Mamba, or a masked-attention encoder trained by self-supervision on the monthly INFORM panel) would produce an embedding $z(u,t) \in \mathbb{R}^d$ that the latent-variable model can consume alongside a. The latent θ stays scalar and interpretable; the encoder has a crisp objective (reduce posterior variance on θ); each learned dimension is registered as a Level-6 property with declared training data and checkpoint hash.

The discipline carries through every extension: declared formulas, declared failure modes, posterior medians always reported with credible intervals, the system as decision support rather than

automation.

10 Conclusion

When almost every active response is underfunded, a single coverage ratio stops discriminating. Treating overlookedness as a latent scalar and inferring its posterior from six observations through a hierarchical Bayesian model gives every ranking an honest credible interval, encodes stakeholder disagreement as a difference between priors over slopes rather than weights in a sum, and is fast enough ($\sim$5 s SVI, $\sim$15 s NUTS) to validate routinely. Implemented over a typed semantic layer so every displayed value traces to source data. Externally validated against two human-curated benchmarks — 5/10 precision @ 10 against CERF UFE 2025 w1 (tied with the additive baseline), 2/7 on CERF UFE 2025 w2 and 3/10 on CARE BTS 2024 (each beating the additive baseline 2×–3×). The variational posterior is calibrated against NUTS on every fit. What remains to be built is the temporal AR(1) extension, the four stakeholder-specific posteriors, and the learned representation layer above the ontology, all held to the same interpretability and provenance discipline.

References

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